

SEO PERFORMANCE ANALYSIS PORTFOLIO

SEO Performance Analysis: Organic Traffic & Keyword Insights

Prepared by:

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Tools & Technologies:

SQL | SQLite| Excel| GitHub

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1. Summary

This project analyses organic search performance data to identify trends, high-performing content, and optimization opportunities. Using SQL and Excel visualizations, the analysis highlights how traffic is distributed across pages, keywords, and regions, as well as the relationship between search rankings and user engagement.

The findings show that a small number of pages drive the majority of clicks, while several high-impression keywords underperform in click-through rates, presenting clear opportunities for optimization. Additionally, organic traffic demonstrates a stable upward trend, and higher search rankings are strongly correlated with improved CTR.

2. Business / Analytical Question

Websites rely heavily on organic search traffic for growth. However, not all pages or keywords perform equally. Which pages, keywords, and traffic factors drive the strongest organic performance, and where are the optimization opportunities to improve SEO visibility and engagement?

This project analyses SEO performance data to identify:

1. High-performing pages and keywords
2. Traffic trends over time
3. Opportunities to improve organic visibility and engagement

3. Dataset

The project uses a dataset structured similarly to Google Search Console-style SEO data for analysis. The synthetic SEO dataset contents include:

- Date
- Page URL
- Keyword
- Clicks
- Impressions
- CTR
- Average Position
- Device
- Country

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4. Tools Used

- SQL (SQLite) – data extraction and aggregation
- Excel – data visualization and charting
- Dataset – synthetic SEO performance data

5. Process (What I Did)

1. Imported and cleaned SEO dataset in SQLite (ensuring no whitespace issues)
2. Used SQL to extract insights:
 - Top-performing pages by clicks
 - Keywords with high impressions and low CTR
 - Traffic trends over time
 - CTR vs search ranking relationship
 - Device and country performance segmentation
3. Exported query results to CSV
4. Built professional visualizations in Excel
5. Interpreted results and translated findings into actionable SEO insights

6. Key Visualizations

The analysis includes the following visualizations and observations:

1. Organic Traffic Performance by Country
 - Shows UK dominating overall traffic volume, followed by US, AU and CA
2. High-Impression Keywords with Low CTR
 - Highlights keywords such as:
 - Free phonics sheets
 - Math activity sheets
 - Indicating strong visibility but poor engagement.
3. CTR vs Search Position
 - Demonstrates inverse relationship between ranking and CTR
 - Higher-ranked positions achieve significantly better click rates.
4. Organic Traffic Trends Over Time
 - Shows consistent organic growth with fluctuations across weeks.

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5. Total Organic Clicks (Short-term Trend)

- Highlights daily variations and short-term performance changes.

7. Key Insights

1. A small number of pages contribute disproportionately to total organic traffic → content concentration
2. Keywords with high impressions but low CTR indicate missed opportunities → strong candidates for optimization
3. Clear relationship between search ranking and CTR → visibility drives user engagement
4. Organic traffic shows steady growth with periodic fluctuations → consistent SEO performance
5. UK drives most of the traffic, suggesting a strong regional concentration.

8. Business Implications and Recommendations (Why This Project Matters)

- Improving low-CTR keywords could significantly increase traffic without new content creation
- Focusing on high-performing content themes can maximize ROI on SEO efforts
- Optimizing rankings has a direct impact on user engagement and traffic acquisition
- Regional traffic insights can guide localization and marketing strategy
- Data-driven SEO decisions can improve both visibility and user conversion potential.

Estimated impact:

- *Improving CTR for high-impression keywords by 5–10% could significantly increase organic traffic without additional content creation.*

9. Challenges and How to Solve Them

Challenge: The synthetic dataset lacks real-world noise and complexity

Solution: In real scenarios, combine data from GA4 and Search Console for richer insights

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Challenge: Limited keyword diversity

Solution: Integrate additional keyword datasets or scrape SERP data

Challenge: No direct conversion metrics

Solution: Extend analysis to include conversion tracking data

10. What I Would Improve Next

To make the project even stronger in a production setting, I would:

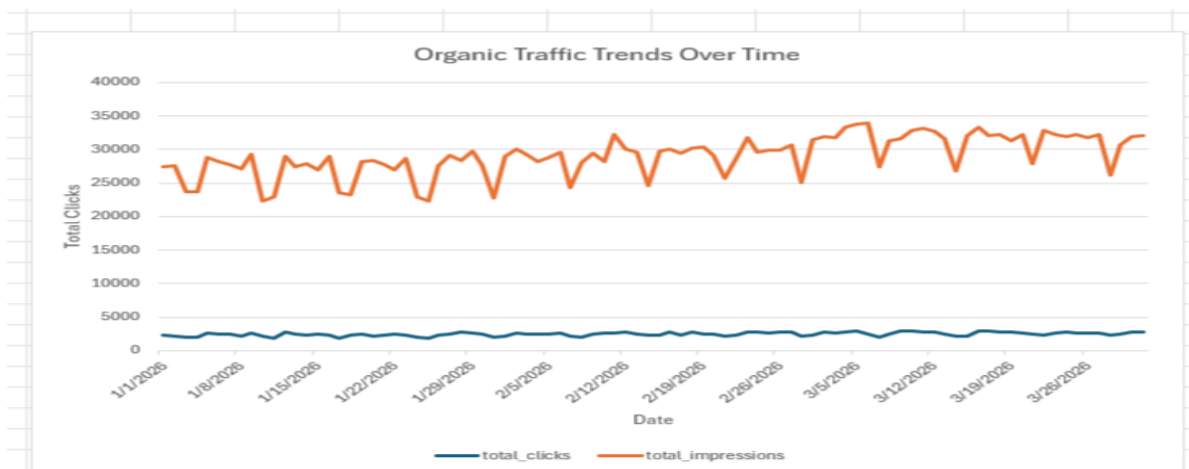
- Integrate real SEO data (GA4 + Search Console)
- Build dashboard in Power BI or Looker Studio
- Perform deeper keyword clustering analysis
- Add A/B testing simulation for SEO changes
- Include conversion and engagement metrics

11. Files and Links

Include in repo:

- SQL queries
- SEO dataset (CSV)
- Excel charts (visual analysis)
- Project report (this document)

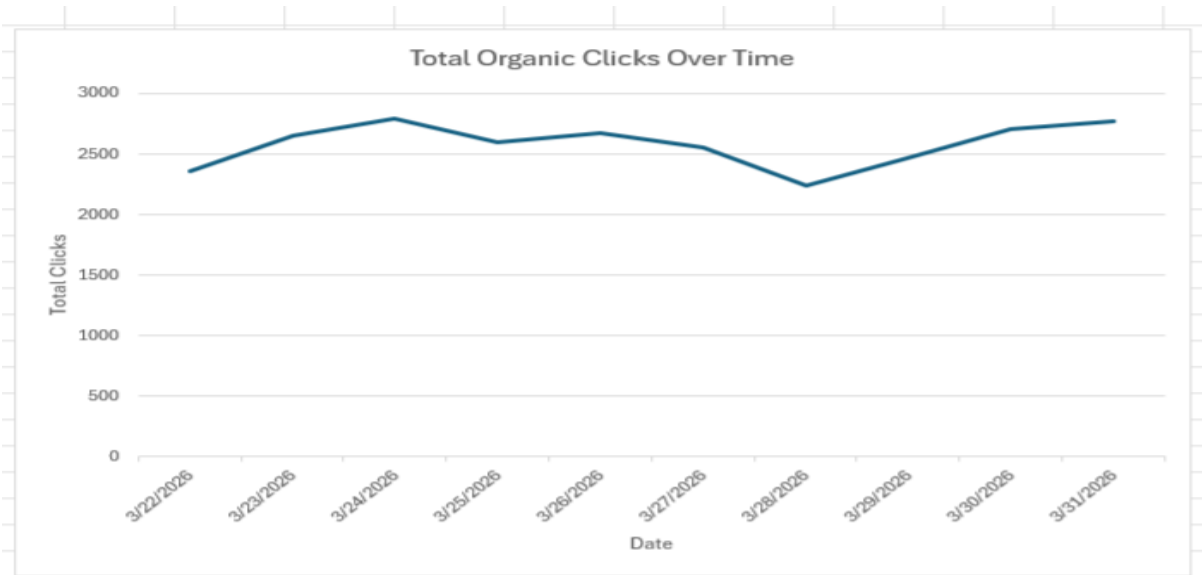
Chart 1: Organic Traffic Trends Over Time



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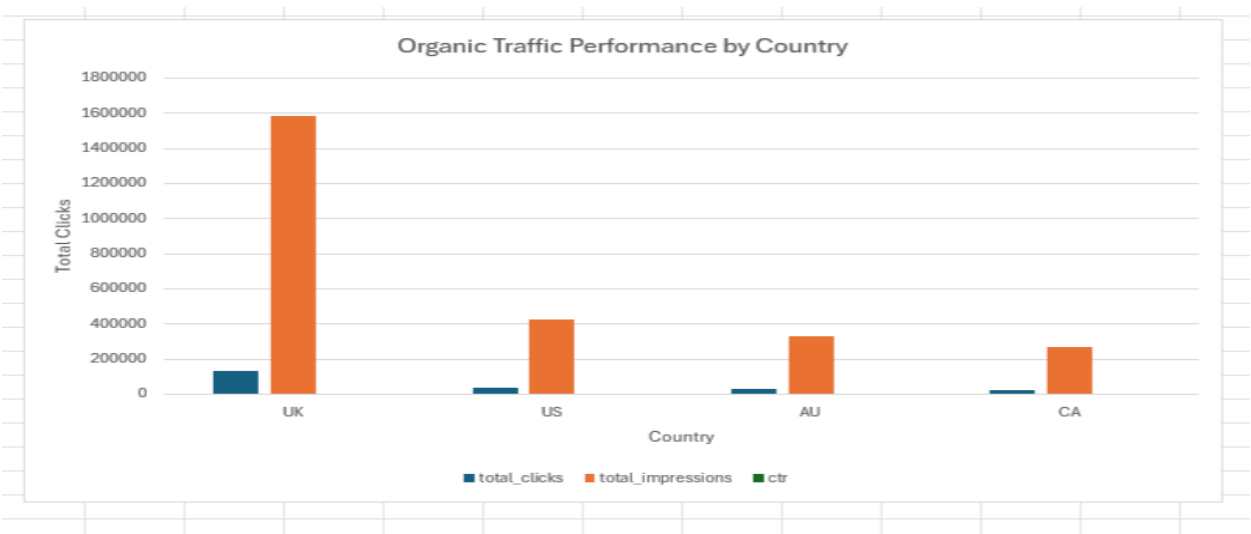
Insight: Organic traffic shows a gradual upward trend over time, indicating consistent SEO performance growth with minor fluctuations.

Chart 2: Total Organic Clicks Over Time



Insight: Daily clicks fluctuate slightly but remain stable overall, suggesting consistent user engagement patterns.

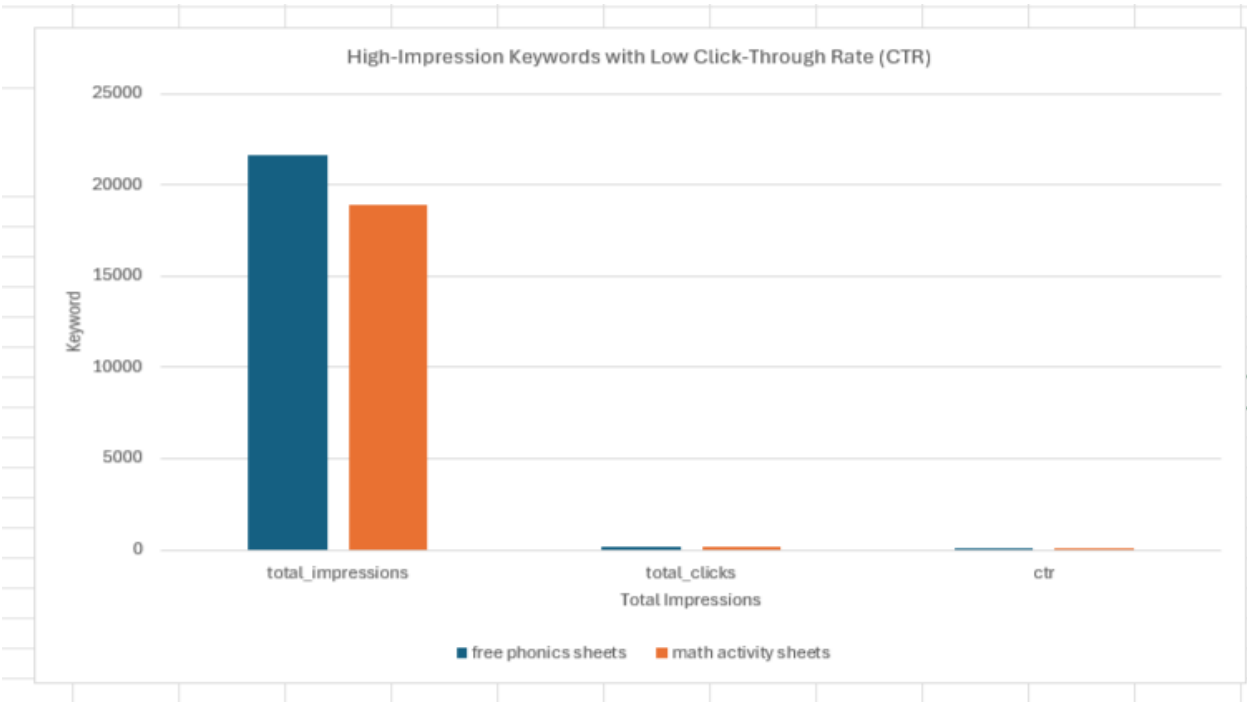
Chart 3: Organic Traffic Performance by Country



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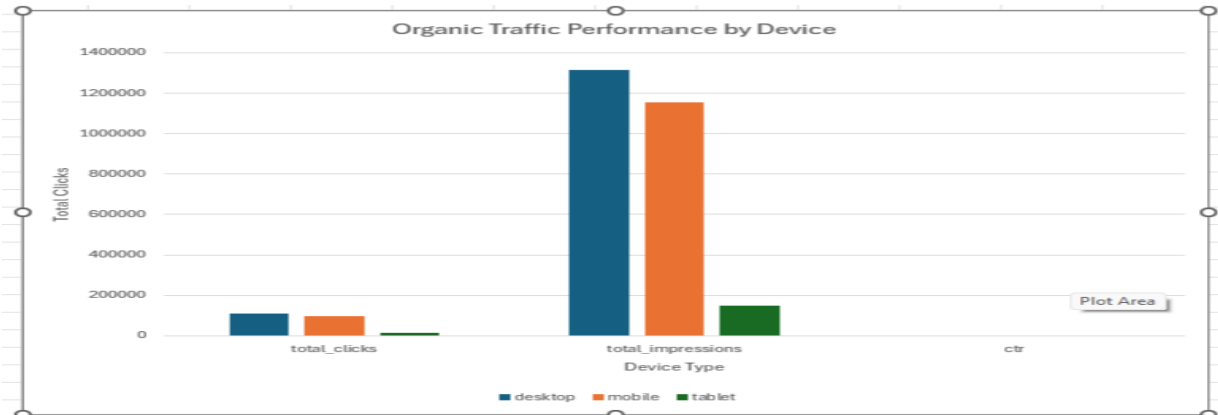
Insight: The UK contributes the majority of traffic, indicating a strong regional concentration and potential focus area for further SEO optimization.

Chart 4: High-impression Keyword with Low Click-through Rate



Insight: Keywords such as “free phonics sheets” and “math activity sheets” generate high impressions but low CTR, highlighting missed optimization opportunities.

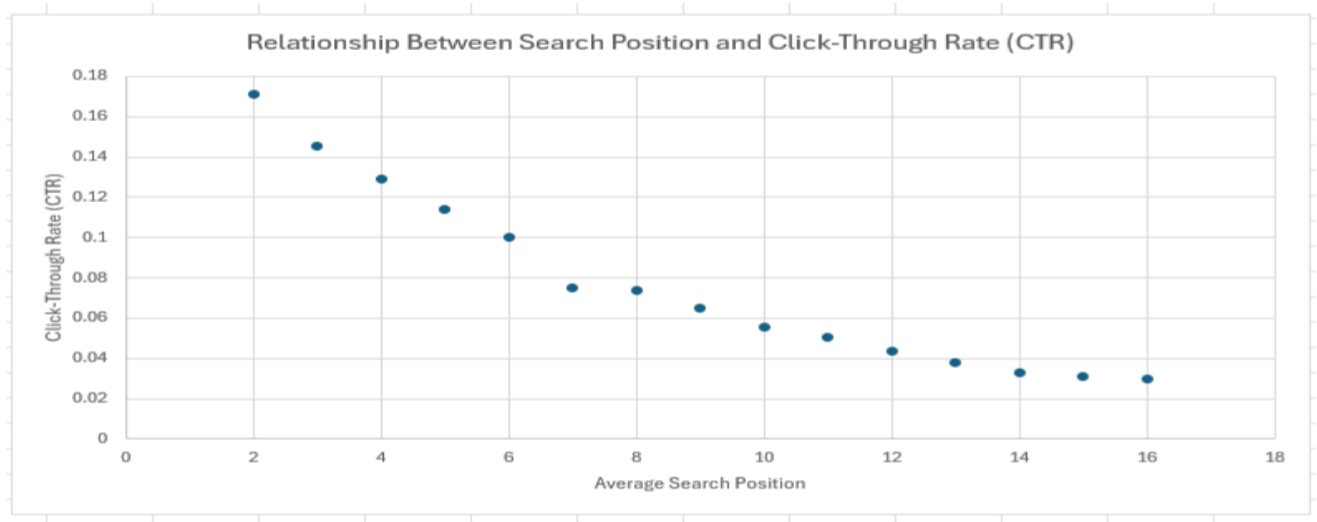
Chart 5: Organic Traffic Performance by Device



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Insight: Mobile and desktop drive most traffic, suggesting the need for device-specific optimization strategies.

Chart 6: Relationship Between Search Position and Click-Through Rate



Insight: Higher-ranking positions result in significantly higher CTR, confirming the importance of improving search visibility.